

Given Single neuron

$$x_1 = 1$$

$$x_2 = 1$$

$$y = 1$$

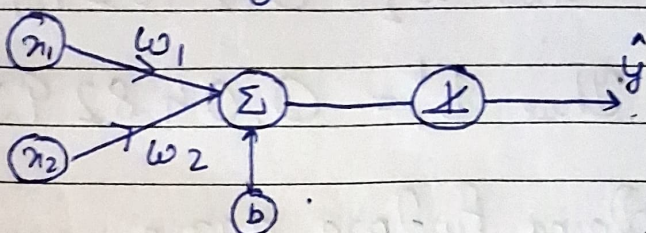
$$w_1 = w_2 = 0.2, \quad b = 0$$

$$\eta = 0.5$$

Activation function is sigmoid

Thus:

The fwd prop



$$\begin{aligned} \text{Thus } z &= w_1 x_1 + w_2 x_2 + b \\ &= 0.2 \times 1 + 0.2 \times 1 + 0 \\ &= 0.4 \end{aligned}$$

$$\text{Thus } \hat{y} = \sigma(0.4) = \frac{1}{1 + e^{-0.4}} = 0.598688$$

$$\text{Thus error } (e) = y - \hat{y}$$

$$= 1 - 0.598688 = 0.401312$$

$$\text{Now } L_0 = \frac{1}{2} (e)^2$$

$$= \frac{1}{2} \times 0.401312^2 = 0.080526$$

$$\begin{aligned} \text{Thus } \frac{\partial L}{\partial w_1} &= \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial w_1} \\ &= (-0.401312) (0.240261) \eta \\ &= -0.096586 \end{aligned}$$

$$\text{Thus } \frac{\partial L}{\partial w_2} = -0.096586$$

$$\text{Thus } \frac{\partial L}{\partial b} = -0.096586$$

$$\begin{aligned} \frac{\partial L}{\partial \hat{y}} &= \frac{d}{d\hat{y}} \left[\frac{1}{2} (y - \hat{y})^2 \right] \\ &= \hat{y} - y \\ &= -0.401312 \end{aligned}$$

$$\begin{aligned} \frac{\partial \hat{y}}{\partial z} &= \frac{d}{dz} (\sigma(z)) \\ &= 0.240261 \end{aligned}$$

$$\omega_{\text{new}} = \omega_{\text{old}} - \eta \frac{\partial L}{\partial \omega}$$

$$\text{Thus } \omega'_1 = 0.2 + 0.5 \times 0.248293 \\ = 0.248293$$

$$\text{Thus } \omega'_2 = 0.248293$$

$$\text{Thus } b'_1 = 0.048293$$

Doing fwdpro again

$$\omega'_1 = \omega'_2 = 0.248293 \\ b' = 0.048293$$

$$\text{Thus } z' = 0.544879$$

$$\text{Thus } \hat{y}' = \frac{1}{1 + e^{-0.544879}} \\ = 0.633 \text{ (Ans)}$$